

Capstone 3: Final Report

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Background:

In the year 2020, the world shut down due to the Covid-19 virus, and from there the world was forever changed. One such change is a growing awareness of how we all approach our health. This was most prevalent in the forms of wearing a facial mask to prevent the spread of illness. There were many different opinions on this topic with people for and being against the mask. However, when it comes to certain places like hospitals, doctors, clinics; the mask can be the difference between life and death. Thus this is the reason why I wanted to focus on this problem as by being able to automatically determine if a person is wearing a facial mask or not, we can help prevent illness from spreading among high risk populations.

Overview / Results:

Using the image repository derived from the MFR2 dataset, this consisted of 269 different images of various celebrities wearing and not wearing masks. The way the images were structured, they were consistent 160x160 size rgb images which helped streamline the process. In terms of other properties, the images had no major issues with brightness, however there are possible issues among the images regarding blurriness. This is likely to be related to the image being smooth rather than being blurry after auditing the most blurriest images.

When it comes to creating the model, this process focuses on using Keras to create an image pipeline. This convolutional neural network (CNN) is created to perform the following: take the input images, perform augmentation on them, then learn the finer details on them across 2 convolutions. After which the model reduces the amount of data the CNN learned about and further processes it into determining if the image has a mask on or not.

The end result however shows that there is more work that can be done. The first model that was created only produced around 60% accuracy. The next step was taking this model and

trying to fine tune it to help increase performance, however the result was still the same. I believe that the most probable reason for this issue is that the overall pipeline is needing improvement as when using an imported model to look at this dataset. It scored a higher accuracy score. Therefore, it just shows that more work needs to be done with the pipeline since by increasing the number of possible answers from 2 to 10, the accuracy also unexplainably went up to around 80%. Hence more research on optimizing the pipeline should be performed.

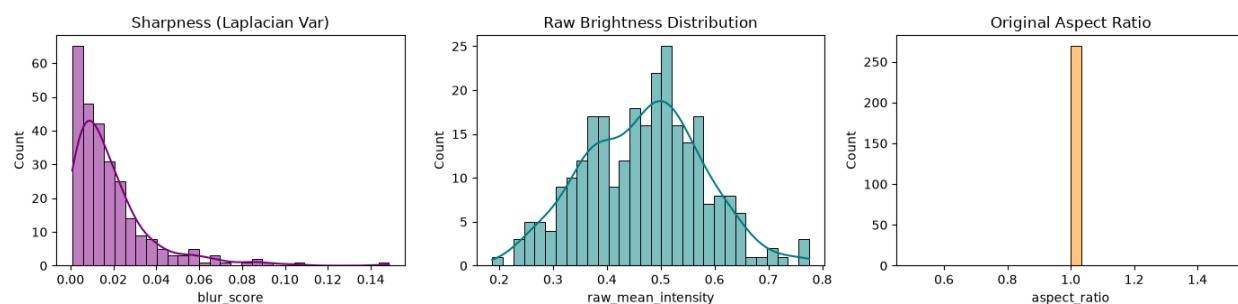


Figure 1: Image properties



Figure 2: blurriest pictures

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Epoch 1/20
c:\Users\Evan\anaconda3\envs\Capstone3\Lib\site-packages\keras\src\trainers\epoch_iterator.py:74: UserWarning: `shuffle=True` was passed, but will be ignored since the data `x` was provided as a tf.
self.data_adapter = data_adapters.get_data_adapter(
c:\Users\Evan\anaconda3\envs\Capstone3\Lib\site-packages\keras\src\ops\nn.py:959: UserWarning: You are using a softmax over axis -1 of a tensor of shape (None, 1). This axis has size 1. The softmax
warnings.warn(
6/6 ————— 4s 420ms/step - accuracy: 0.6561 - loss: 0.6697 - val_accuracy: 0.5875 - val_loss: 0.6883
Epoch 2/20
6/6 ————— 2s 418ms/step - accuracy: 0.6561 - loss: 0.5467 - val_accuracy: 0.5875 - val_loss: 0.6759
Epoch 3/20
6/6 ————— 2s 410ms/step - accuracy: 0.6561 - loss: 0.5170 - val_accuracy: 0.5875 - val_loss: 0.6735
Epoch 4/20
6/6 ————— 3s 429ms/step - accuracy: 0.6561 - loss: 0.4805 - val_accuracy: 0.5875 - val_loss: 0.6677
Epoch 5/20
6/6 ————— 2s 412ms/step - accuracy: 0.6561 - loss: 0.4597 - val_accuracy: 0.5875 - val_loss: 0.6671
Epoch 6/20
6/6 ————— 2s 391ms/step - accuracy: 0.6561 - loss: 0.4242 - val_accuracy: 0.5875 - val_loss: 0.6651
Epoch 7/20
6/6 ————— 2s 395ms/step - accuracy: 0.6561 - loss: 0.4245 - val_accuracy: 0.5875 - val_loss: 0.6660
Epoch 8/20
6/6 ————— 2s 390ms/step - accuracy: 0.6561 - loss: 0.4153 - val_accuracy: 0.5875 - val_loss: 0.6628
Epoch 9/20
6/6 ————— 2s 387ms/step - accuracy: 0.6561 - loss: 0.3839 - val_accuracy: 0.5875 - val_loss: 0.6629
Epoch 10/20
6/6 ————— 2s 385ms/step - accuracy: 0.6561 - loss: 0.3519 - val_accuracy: 0.5875 - val_loss: 0.6643
Epoch 11/20
6/6 ————— 2s 383ms/step - accuracy: 0.6561 - loss: 0.3795 - val_accuracy: 0.5875 - val_loss: 0.6675
Epoch 12/20
6/6 ————— 2s 388ms/step - accuracy: 0.6561 - loss: 0.3622 - val_accuracy: 0.5875 - val_loss: 0.6707
Epoch 13/20
6/6 ————— 2s 413ms/step - accuracy: 0.6561 - loss: 0.3343 - val_accuracy: 0.5875 - val_loss: 0.6901
...
Epoch 19/20
6/6 ————— 2s 391ms/step - accuracy: 0.6561 - loss: 0.3243 - val_accuracy: 0.5875 - val_loss: 0.9262
Epoch 20/20
6/6 ————— 2s 386ms/step - accuracy: 0.6561 - loss: 0.2786 - val_accuracy: 0.5875 - val_loss: 0.9757
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

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Figure 3: Initial model results

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Epoch 1/30
6/6 ————— 3s 420ms/step - accuracy: 0.6561 - loss: 0.3208 - val_accuracy: 0.5875 - val_loss: 0.9533
Epoch 2/30
6/6 ————— 2s 405ms/step - accuracy: 0.6561 - loss: 0.3047 - val_accuracy: 0.5875 - val_loss: 1.0640
Epoch 3/30
6/6 ————— 2s 410ms/step - accuracy: 0.6561 - loss: 0.3088 - val_accuracy: 0.5875 - val_loss: 1.1074
Epoch 4/30
6/6 ————— 2s 408ms/step - accuracy: 0.6561 - loss: 0.3057 - val_accuracy: 0.5875 - val_loss: 1.1082
Epoch 5/30
6/6 ————— 2s 404ms/step - accuracy: 0.6561 - loss: 0.2610 - val_accuracy: 0.5875 - val_loss: 1.2833
Epoch 6/30
6/6 ————— 2s 405ms/step - accuracy: 0.6561 - loss: 0.2623 - val_accuracy: 0.5875 - val_loss: 1.2832
Epoch 7/30
6/6 ————— 2s 405ms/step - accuracy: 0.6561 - loss: 0.2343 - val_accuracy: 0.5875 - val_loss: 1.5122
Epoch 8/30
6/6 ————— 2s 410ms/step - accuracy: 0.6561 - loss: 0.2311 - val_accuracy: 0.5875 - val_loss: 1.4734
Epoch 9/30
6/6 ————— 2s 393ms/step - accuracy: 0.6561 - loss: 0.2055 - val_accuracy: 0.5875 - val_loss: 1.7093
Epoch 10/30
6/6 ————— 2s 402ms/step - accuracy: 0.6561 - loss: 0.1781 - val_accuracy: 0.5875 - val_loss: 1.6045
Epoch 11/30
6/6 ————— 2s 394ms/step - accuracy: 0.6561 - loss: 0.2243 - val_accuracy: 0.5875 - val_loss: 1.8290
Epoch 12/30
6/6 ————— 2s 404ms/step - accuracy: 0.6561 - loss: 0.1985 - val_accuracy: 0.5875 - val_loss: 2.0616
Epoch 13/30
...
Epoch 29/30
6/6 ————— 2s 383ms/step - accuracy: 0.6561 - loss: 0.1474 - val_accuracy: 0.5875 - val_loss: 3.5818
Epoch 30/30
6/6 ————— 2s 405ms/step - accuracy: 0.6561 - loss: 0.1329 - val_accuracy: 0.5875 - val_loss: 2.8398

```

Figure 4: Tuned Model results

Validation Accuracy: 0.8875				
Classification Report:				
	precision	recall	f1-score	support
0	1.00	0.73	0.84	33
1	0.84	1.00	0.91	47
accuracy			0.89	80
macro avg	0.92	0.86	0.88	80
weighted avg	0.91	0.89	0.88	80

Figure 5: Imported Model Results

Recommendations

Overall I recommend that this project be continued in terms of refinement as it shows that there are working models out there that fulfill the purpose. As for other recommendations, I believe that outside of more images for better variety in further training and testing the model; there are not many more recommendations I can give. One thing that we need to be aware of is that since human faces can be varied, there can be inherent bias. Although this model is only checking if a subject is wearing a mask which would appear to be different than a face, it should be considered as well.